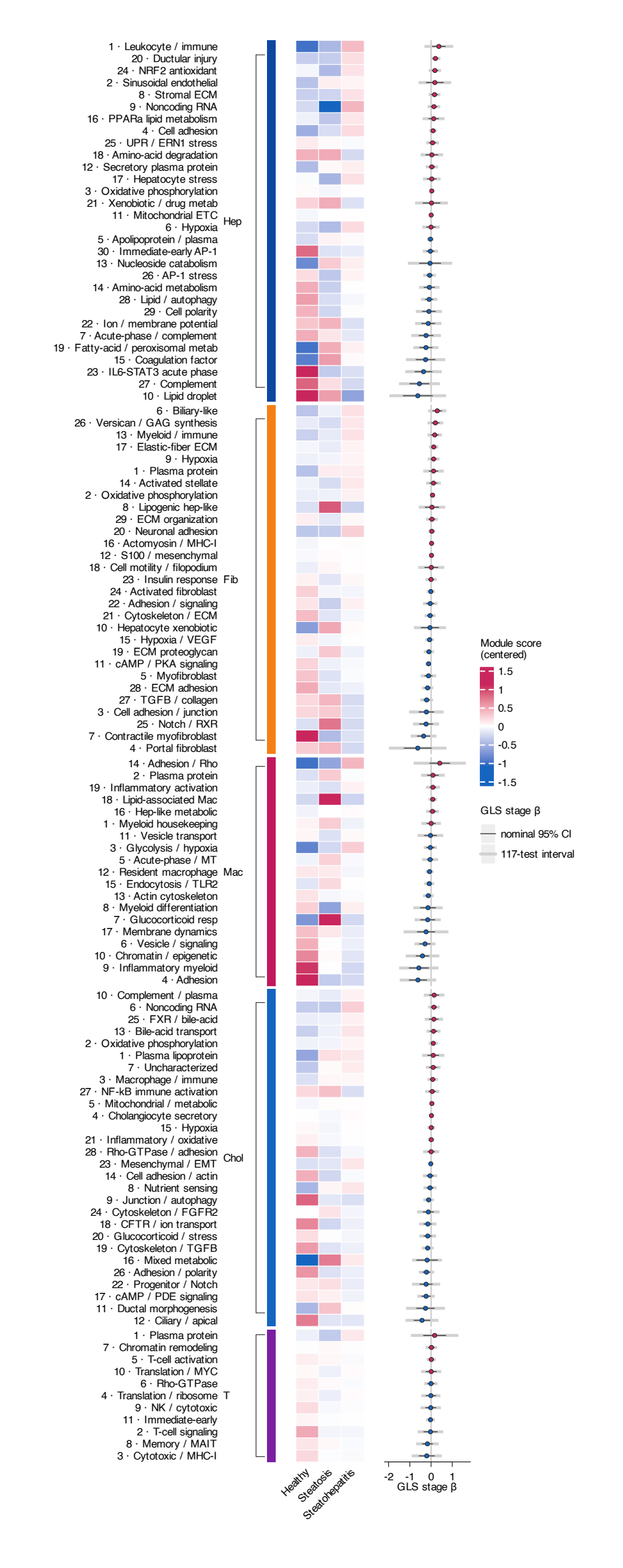
1 · Leukocyte / immune
20 · Ductular injury
24 · NRF2 antioxidant
2 · Sinusoidal endothelial
8 · Stromal ECM
9 · Noncoding RNA
16 · PPARa lipid metabolism
4 · Cell adhesion
25 · UPR / ERN1 stress
18 · Amino-acid degradation
12 · Secretory plasma protein
17 · Hepatocyte stress
3 · Oxidative phosphorylation
21 · Xenobiotic / drug metab
11 · Mitochondrial ETC
6 · Hypoxia
5 · Apolipoprotein / plasma
30 · Immediate-early AP-1
13 · Nucleoside catabolism
26 · AP-1 stress
14 · Amino-acid metabolism
28 · Lipid / autophagy
29 · Cell polarity
22 · Ion / membrane potential
7 · Acute-phase / complement
19 · Fatty-acid / peroxisomal metab
15 · Coagulation factor
23 · IL6-STAT3 acute phase
27 · Complement
10 · Lipid droplet
Hep
6 · Biliary-like
26 · Versican / GAG synthesis
13 · Myeloid / immune
17 · Elastic-fiber ECM
9 · Hypoxia
1 · Plasma protein
14 · Activated stellate
2 · Oxidative phosphorylation
8 · Lipogenic hep-like
29 · ECM organization
20 · Neuronal adhesion
16 · Actomyosin / MHC-I
12 · S100 / mesenchymal
18 · Cell motility / filopodium
23 · Insulin response
24 · Activated fibroblast
22 · Adhesion / signaling
21 · Cytoskeleton / ECM
10 · Hepatocyte xenobiotic
15 · Hypoxia / VEGF
19 · ECM proteoglycan
11 · cAMP / PKA signaling
5 · Myofibroblast
28 · ECM adhesion
27 · TGFB / collagen
3 · Cell adhesion / junction
25 · Notch / RXR
7 · Contractile myofibroblast
4 · Portal fibroblast
Fib
14 · Adhesion / Rho
2 · Plasma protein
19 · Inflammatory activation
18 · Lipid-associated Mac
16 · Hep-like metabolic
1 · Myeloid housekeeping
11 · Vesicle transport
3 · Glycolysis / hypoxia
5 · Acute-phase / MT
12 · Resident macrophage
15 · Endocytosis / TLR2
13 · Actin cytoskeleton
8 · Myeloid differentiation
7 · Glucocorticoid resp
17 · Membrane dynamics
6 · Vesicle / signaling
10 · Chromatin / epigenetic
9 · Inflammatory myeloid
4 · Adhesion
Mac
10 · Complement / plasma
6 · Noncoding RNA
25 · FXR / bile-acid
13 · Bile-acid transport
2 · Oxidative phosphorylation
1 · Plasma lipoprotein
7 · Uncharacterized
3 · Macrophage / immune
27 · NF-kB immune activation
5 · Mitochondrial / metabolic
4 · Cholangiocyte secretory
15 · Hypoxia
21 · Inflammatory / oxidative
28 · Rho-GTPase / adhesion
23 · Mesenchymal / EMT
14 · Cell adhesion / actin
8 · Nutrient sensing
9 · Junction / autophagy
24 · Cytoskeleton / FGFR2
18 · CFTR / ion transport
20 · Glucocorticoid / stress
19 · Cytoskeleton / TGFB
16 · Mixed metabolic
26 · Adhesion / polarity
22 · Progenitor / Notch
17 · cAMP / PDE signaling
11 · Ductal morphogenesis
12 · Ciliary / apical
Chol
1 · Plasma protein
7 · Chromatin remodeling
5 · T-cell activation
10 · Translation / MYC
6 · Rho-GTPase
4 · Translation / ribosome
9 · NK / cytotoxic
11 · Immediate-early
2 · T-cell signaling
8 · Memory / MAIT
3 · Cytotoxic / MHC-I
T
Healthy
Steatosis
Steatohepatitis
-2
-1
0
1
GLS stage β
Module score (centered)
1.5
1
0.5
0
-0.5
-1
-1.5
GLS stage β
nominal 95% CI
117-test interval